ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 4 · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics practice exam 4.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, equivalent designs and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 A ✓ 1.1.3 D ✓ 1.1.4 D ✓ 1.1.5 B ✓ 1.1.6 C ✓ 1.1.7 C ✓ 1.1.8 A ✓ 1.1.9 A ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 C ✓ 1.2.2 I ✓ 1.2.3 G ✓ 1.2.4 E ✓ 1.2.5 J ✓ 1.2.6 A ✓ 1.2.7 F ✓ 1.2.8 D ✓ 1.2.9 H ✓ 1.2.10 B ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1 Blocks in order (230 V AC → transformer → rectifier → filter → regulator → 5 V DC load) ✓ with functions ✓✓✓✓: transformer – steps 230 V down to a lower AC voltage; rectifier – changes AC into pulsating DC; filter – smooths the ripple; regulator – keeps the output at a steady 5 V. (5)

2.2 It uses both half-cycles of the AC ✓, so the output has less ripple / is easier to smooth and the transformer is used more efficiently ✓. (2)

2.3 The capacitor charges to the peak voltage ✓ and discharges slowly into the load between peaks, filling in the gaps so that the voltage stays close to the peak ✓. (2)

2.4.1 $I = 20 + 10 = 30\ mA$ ✓; $R = (V_s - V_z) \div I = (9 - 5{,}1) \div 0{,}03$ ✓ $= 130\ \Omega$ ✓ (3)

2.4.2 $P = I^2R$ ✓ $= 0{,}03^2 \times 130 = 0{,}12\ W$ ✓ (2)

2.5 $S = V \times I$ ✓ $= 12 \times 1{,}5 = 18\ VA$ ✓ (2)

2.6.1 Parallel: $V = 18\ V, I = 3 + 3 = 6\ A$ ✓; $P = 18 \times 6 = 108\ W$ ✓ (2)

2.6.2 $t = Ah \div I$ ✓ $= 20 \div 5 = 4\ h$ ✓ (2)

[20]

QUESTION 3

3.1.1 $V_{out} = V \times R \div (R + R_{LDR})$ ✓. Dark: $5 \times 10 \div (10 + 100) = 0{,}45\ V$ ✓✓. Bright: $5 \times 10 \div (10 + 1) = 4{,}55\ V$ ✓ (4)

3.1.2 It rises ✓ (1)

3.2 $R = (5 - 2) \div 0{,}02$ ✓ $= 150\ \Omega$ ✓; $P = 0{,}02^2 \times 150 = 0{,}06\ W$ ✓ (3)

3.3 Marks ✓✓✓✓✓✓: microcontroller pin to a base resistor; base resistor to the base of the NPN transistor; emitter to 0 V (common ground with the microcontroller); relay coil between +12 V and the collector; flywheel diode across the coil, cathode to +12 V; neat drawing with correct symbols and labels. (6)

3.4 Any FOUR ✓✓✓✓: check for solder bridges between tracks; check that the copper tracks were cut in the right places; check components are in the right holes and the right way round (transistor pins, diode); check for dry joints; measure the supply voltage on the board; check continuity of tracks and wire links; measure the base voltage when the pin is HIGH. (4)

3.5 The sensor voltage goes to one input and a reference voltage (voltage divider) to the other ✓; when the sensor voltage rises above the reference the output swings HIGH (saturates) and switches on the fan through a transistor / relay ✓. (2)

[20]

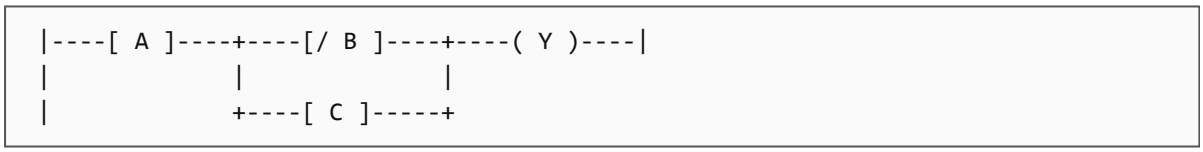
QUESTION 4

4.1.1 ABC → Y: 000→0, 001→0, 010→0, 011→0, 100→1, 101→1, 110→0, 111→1. Inputs correct ✓; A = 0 rows all 0 ✓; 100 and 101 = 1 ✓; 110 = 0 and 111 = 1 ✓ (4)

4.1.2 $Y = A \cdot (\bar{B} + C)$ ✓✓ $(\text{or } Y = A\bar{B} + AC)$ (2)

4.1.3 NOT gate on B ✓; OR gate with inputs $\bar{B}$ and C ✓; AND gate with inputs A and the OR output ✓; output labelled Y, neat ✓ (equivalent AND-OR circuit for $A\bar{B}$ + AC accepted) (4)

4.2 Rung: normally open contact A ✓ in series with a parallel branch ✓ of a normally closed contact B ✓ and a normally open contact C, driving output coil Y ✓, e.g. (4)



4.3 Join both inputs of the NAND gate together (or tie one input HIGH); the output is then $\bar{A}$ ✓. Any basic gate (AND, OR, NOT) can be built from NAND gates only ✓. (2)

4.4 Current in the coil creates a magnetic field ✓ that pulls the soft-iron plunger into the coil, moving it in a straight line; a spring returns it when the current stops ✓. (2)

4.5 Any TWO ✓✓: more turns on the coil; a larger current / higher voltage; a better (higher permeability) iron core or plunger; a smaller air gap. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF MARKING GUIDELINE